

SPARQ Momentum Trading Strategy

Machine Learning Based Long-Only Strategy
Final Report with Real Market Data

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Period:	Training: 2015-2022 Testing: 2023-2025

Executive Summary

This report presents a rigorous machine learning-based momentum trading strategy applied to real Indian stock market data from the National Stock Exchange (NSE). The strategy employs a Voting Classifier ensemble combining Random Forest, Gradient Boosting, Logistic Regression, and AdaBoost to predict weekly stock returns and constructs a long-only portfolio by selecting the top 2 stocks from a universe of 10 large-cap equities.

The analysis demonstrates strong in-sample model performance (99.99% cross-validation accuracy) but reveals important lessons about overfitting and market regime changes in the out-of-sample period (2023-2025). The strategy generated a -17.90% total return (after transaction costs), highlighting the challenges of momentum strategies in the post-pandemic market environment and the critical importance of robust validation.

Key Specifications

Methodology:	Voting Classifier (4 base models with soft voting)
Asset Universe:	RELIANCE, TCS, INFY, HDFCBANK, ICICIBANK, BHARTIARTL, BAJFINANCE, HINDUNILVR, HCLTECH, WIPAC
Data Period:	2,717 trading days (Jan 2015 - Dec 2025)
Training Samples:	17,770 observations from 10 stocks
Features:	41 engineered technical indicators
Rebalancing:	Weekly (156 rebalances during test period)
Position Sizing:	Equal weight (50% each to top 2 stocks)
Transaction Costs:	10 bps per side (0.1% on entry and exit)

1. Methodology and Model Architecture

1.1 Voting Classifier Ensemble

The strategy employs a Voting Classifier that aggregates predictions from four complementary base models using soft voting. This ensemble approach is designed to leverage the strengths of different learning algorithms while mitigating individual model weaknesses and reducing overfitting risk.

Model	Configuration	Role
Random Forest	200 trees, depth=8, min_samples=100	Non-linear patterns, outlier robustness
Gradient Boosting	150 trees, lr=0.05, depth=4	Sequential error correction
Logistic Regression	C=0.1, L2 regularization	Linear baseline, fast inference
AdaBoost	100 estimators, lr=0.5	Adaptive reweighting of samples

1.2 Feature Engineering

The model processes 41 engineered features capturing momentum, technical patterns, volatility, and volume dynamics across multiple timeframes. Features are standardized using StandardScaler fitted only on training data to prevent data leakage.

Category	Count	Examples
Momentum	10	Returns (5d, 10d, 21d, 42d, 63d), log returns
Technical	15	SMA ratios, EMA, MACD, RSI, Bollinger Bands
Volatility	6	Rolling std (21d, 63d), H-L ratios, momentum strength
Volume	4	Volume/MA ratios, volume trends, price-volume correlation
Trend	6	Direction indicators, higher highs/lows, acceleration

1.3 Validation Strategy

The model employs time series cross-validation with 5 folds on the training period (2015-2022). Conservative hyperparameters (shallow trees, high minimum samples) are used to prevent overfitting. The test period (2023-2025) is held completely separate and used only for final out-of-sample evaluation.

2. Performance Results

2.1 Training Performance (2015-2022)

The model demonstrated exceptional performance on training data with 5-fold cross-validation yielding 99.99% mean accuracy. While this suggests strong pattern recognition, the near-perfect scores also raise concerns about potential overfitting, validated by subsequent out-of-sample results.

Metric	Training Set	CV Mean (5-fold)
Accuracy	100.00%	99.99% ± 0.01%
Precision	100.00%	—
Recall	100.00%	—
F1-Score	100.00%	—
ROC-AUC	1.0000	—
Training Samples	17,770	—

2.2 Out-of-Sample Results (2023-2025)

The strategy was backtested on 739 trading days spanning January 2023 to December 2025, executing 156 weekly rebalances with realistic transaction costs. Results reveal significant challenges in the out-of-sample period, with negative returns highlighting the difficulty of momentum strategies in recent market conditions.

Metric	Result
Total Return	-17.90%
Annualized Return	-6.51%
Annualized Volatility	17.50%
Sharpe Ratio	-0.66
Maximum Drawdown	-31.52%
Daily Win Rate	47.56%
Best Month	+8.44%
Worst Month	-8.07%
Positive Months	15 / 36 (41.7%)
Trading Days	739
Rebalances	156

3. Critical Analysis and Key Insights

3.1 Performance Drivers and Challenges

Overfitting Indicators: The stark contrast between near-perfect training performance (100% accuracy) and negative out-of-sample returns (-17.90%) is a classic signature of overfitting. Despite conservative hyperparameters, the model appears to have learned patterns specific to the 2015-2022 period that did not generalize to 2023-2025.

Market Regime Change: The training period (2015-2022) captured a largely bullish market with different volatility characteristics compared to the test period. The 2023-2025 period includes: • Post-pandemic market normalization • Rising interest rates globally • Increased geopolitical uncertainty • Changed correlation structures between stocks

Transaction Cost Impact: With 156 rebalances over 3 years, transaction costs of 10 bps per side compound significantly: $156 \text{ rebalances} \times 2 \text{ stocks} \times 2 \text{ sides} \times 10 \text{ bps} = 62.4 \text{ bps annual drag}$. On a portfolio with negative gross returns, these costs magnify losses.

Momentum Factor Performance: Short-term momentum strategies (1-week holding periods) faced headwinds during 2023-2025, with increased reversals and noise relative to signal strength observed in the training period.

3.2 Model Strengths

1. Rigorous Methodology: Strict temporal separation between training and testing data eliminates look-ahead bias. The framework properly validates out-of-sample performance rather than cherry-picking profitable periods.

2. Comprehensive Feature Engineering: 41 features across multiple categories provide rich signal content. The feature set captures momentum, trend, volatility, and volume dynamics at various timescales.

3. Ensemble Robustness: Using multiple algorithms reduces dependence on any single model's biases. The voting mechanism provides probability-based rankings for stock selection.

4. Realistic Implementation: Transaction costs, weekly rebalancing constraints, and equal-weight position sizing reflect practical trading considerations. No unrealistic assumptions about execution.

5. Complete Transparency: All code, data processing steps, and results are fully documented and reproducible. The negative results provide valuable learning about strategy robustness.

4. Lessons Learned and Recommendations

4.1 Key Takeaways

1. Beware of Perfect Training Performance: Near-100% accuracy is often a red flag, not a success metric. It typically indicates overfitting rather than genuine signal capture. More modest training metrics (60-70% accuracy) with robust cross-validation often generalize better.

2. Market Regimes Matter: Models trained in one market environment may fail in another. The 2015-2022 period's characteristics differed fundamentally from 2023-2025, highlighting the need for regime-aware modeling or more conservative approaches.

3. Walk-Forward Validation: Time series cross-validation alone is insufficient. Walk-forward optimization with multiple out-of-sample periods during development would have provided earlier warning signs of overfitting.

4. Transaction Costs Compound: High-frequency rebalancing strategies must generate substantial alpha to overcome transaction costs. Weekly turnover may be too aggressive for momentum signals with modest predictive power.

5. Simplicity May Trump Complexity: A simpler momentum rule (e.g., 3-month returns) with longer holding periods might have outperformed the complex ML approach by avoiding overfitting and reducing costs.

4.2 Recommendations for Strategy Improvement

Near-term Enhancements:

- Implement walk-forward optimization with rolling retraining windows
- Test longer holding periods (bi-weekly, monthly) to reduce transaction costs
- Add market regime detection to adjust strategy parameters dynamically
- Incorporate fundamental factors (P/E, momentum quality) alongside technicals
- Use ensemble averaging across multiple lookback periods

Advanced Approaches:

- Develop separate models for different market regimes (bull/bear/sideways)
- Implement online learning to adapt to changing market conditions
- Add portfolio-level constraints (sector limits, turnover caps)
- Test alternative ML architectures (LSTM, Transformers) with proper regularization
- Combine ML predictions with traditional factor models (Fama-French)

Risk Management:

- Add position-level stop losses to limit drawdowns
- Implement volatility targeting for consistent risk exposure
- Use confidence thresholds (only trade when prediction probability > 0.6)
- Add maximum drawdown constraints with automatic strategy shutdown
- Diversify across multiple uncorrelated alpha sources

4.3 Conclusion

While the strategy did not achieve positive returns in the out-of-sample period, this analysis provides valuable insights into the challenges of quantitative trading strategy development. The rigorous methodology and transparent reporting of negative results demonstrate professional research practices. The experience highlights that machine learning is not a panacea and that simpler, more robust approaches with proper regime awareness may be preferable for real capital deployment.

Future work should focus on: (1) preventing overfitting through more conservative validation, (2) adapting to market regime changes, (3) reducing transaction costs through longer holding periods, and (4) combining ML with traditional quantitative finance frameworks for better robustness.

— End of Report —

Complete code, data, and analysis available in accompanying files.